

Problem Set 2

Course Name: Modeling and Control	LP / Year: 4 / 2025	Course Code: R0004E
Number of questions: 7	Release Date: 2025-04-09	Submission Date:

1. (a) Mention and explain the two main approaches to develop a model of a dynamical system.
- (b) Based on observation of the step response, most dynamical systems can be approximated as either a first order or a second order system. Specify the main difference between the first and second order system, based on its response. Please explain using graphical illustration to help.
- (c) Provide the main parameters of each first order and second order transfer functions. Provide an example of each transfer function, and determine the value of the corresponding parameters.

2. Given the pole plot shown in Figure 1.

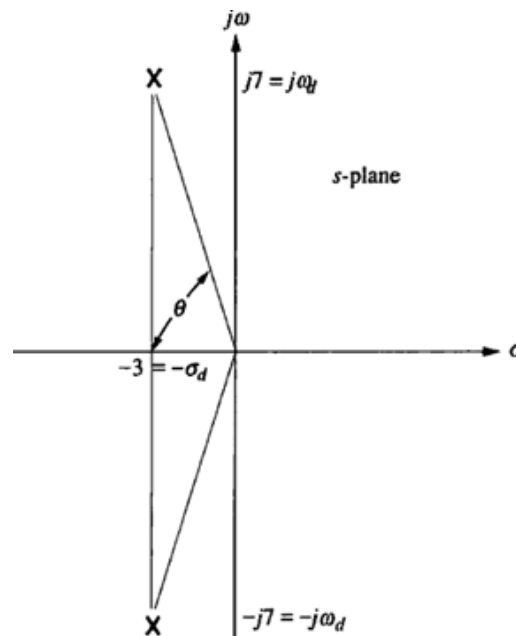


Figure 1: Pole plot of a dynamical system.

- (a) Write the transfer function of the system.
- (b) Find the second order parameters ζ and ω_n of the system.
- (c) Determine the percent overshoot M_p , settling time t_s and rise time t_r of the response with respect to a unit step input.

- (d) Sketch the unit step response of the system by making use of the parameters you obtained in (b) and (c).
- (e) Plot the unit step response using MATLAB, and compare the plot you obtain using MATLAB with your sketch in (d).

3. A control system has a state-space model represented by:

$$\dot{x} = Ax + Bu = \begin{bmatrix} -4 & -1.5 \\ 4 & 0 \end{bmatrix} x + \begin{bmatrix} 2 \\ 0 \end{bmatrix} u; \quad y = Cx + Du = \begin{bmatrix} 1.5 & 0.625 \end{bmatrix} x$$

- (a) Find the eigenvalue of the state matrix A.
- (b) Show that the equivalent transfer function of the system is

$$\frac{Y(s)}{U(s)} = \frac{3s + 5}{s^2 + 4s + 6}$$

Hint: Use the formula $\frac{Y(s)}{U(s)} = C(sI - A)^{-1}B + D$, with s the Laplace variable.

- (c) Find the poles of the system and determine the stability of the system.
- (d) Compare these poles with the eigenvalues that you obtain in (a). Comment on the relation between the stability of the system in a state-space representation and in transfer function representation.
- (e) Use the MATLAB command `tf2ss` to obtain the state-space model of $\frac{Y(s)}{U(s)}$ and compare the result to the original state-space model above.

4. Consider the position control system in Figure 2, in which K is an adjustable gain parameter and

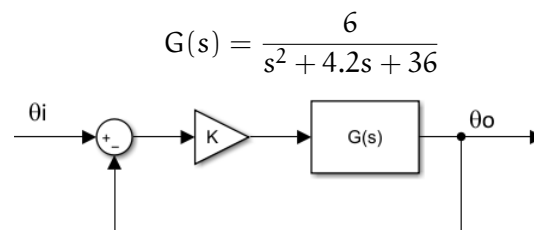


Figure 2: Block diagram of a position control system.

- (a) For $K=1$, find the closed-loop transfer function $T(s)$.
- (b) Find the steady state error of each the open-loop system $G(s)$ and the closed-loop system $T(s)$, with respect to a unit step input ($\Theta_i(s) = \frac{1}{s}$).
- (c) Explain your answer by relating the steady-state errors that you find, to the type of the system.
- (d) Compare the steady state errors that you obtained in (b), which one is larger and give explanation.

5. Repeat Questions 4(a) and (b), for the transfer function:

$$G_1(s) = \frac{s + 6}{s^2 + 4.2s + 36}$$

- (a) Compare the steady state error of the closed-loop system $T_1(s)$, with the one you obtain for $T(s)$ in Question 4(b).
- (b) Explain the effect of a zero in $G_1(s)$ to the closed-loop system $T_1(s)$, taking the comparison with $T(s)$.

6. Given a block diagram of a complex system as shown in Figure 3:

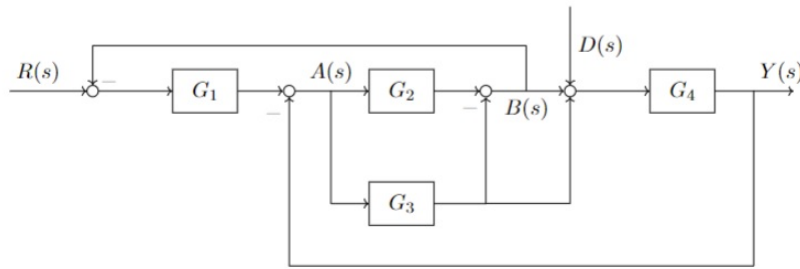


Figure 3: Block diagram of a complex system

- Simplify the block diagram and find the transfer functions $\frac{Y(s)}{R(s)}$.
- Simplify the block diagram and find the transfer functions $\frac{Y(s)}{D(s)}$.

7. The block diagram of a control system is depicted in Figure 4.

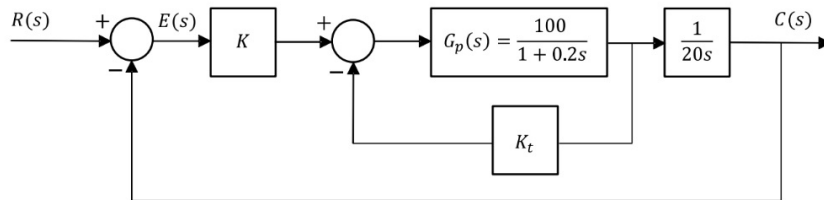


Figure 4: Block diagram of a double loop system

- Determine the stability of the closed-loop system if $K = 1$ and $K_t = 0$.
- Find the steady-state error in unit-ramp response ($r(t) = t$) in terms of K and K_t .
- Discuss the effects that varying the values of K and K_t has on the steady-state error.

END